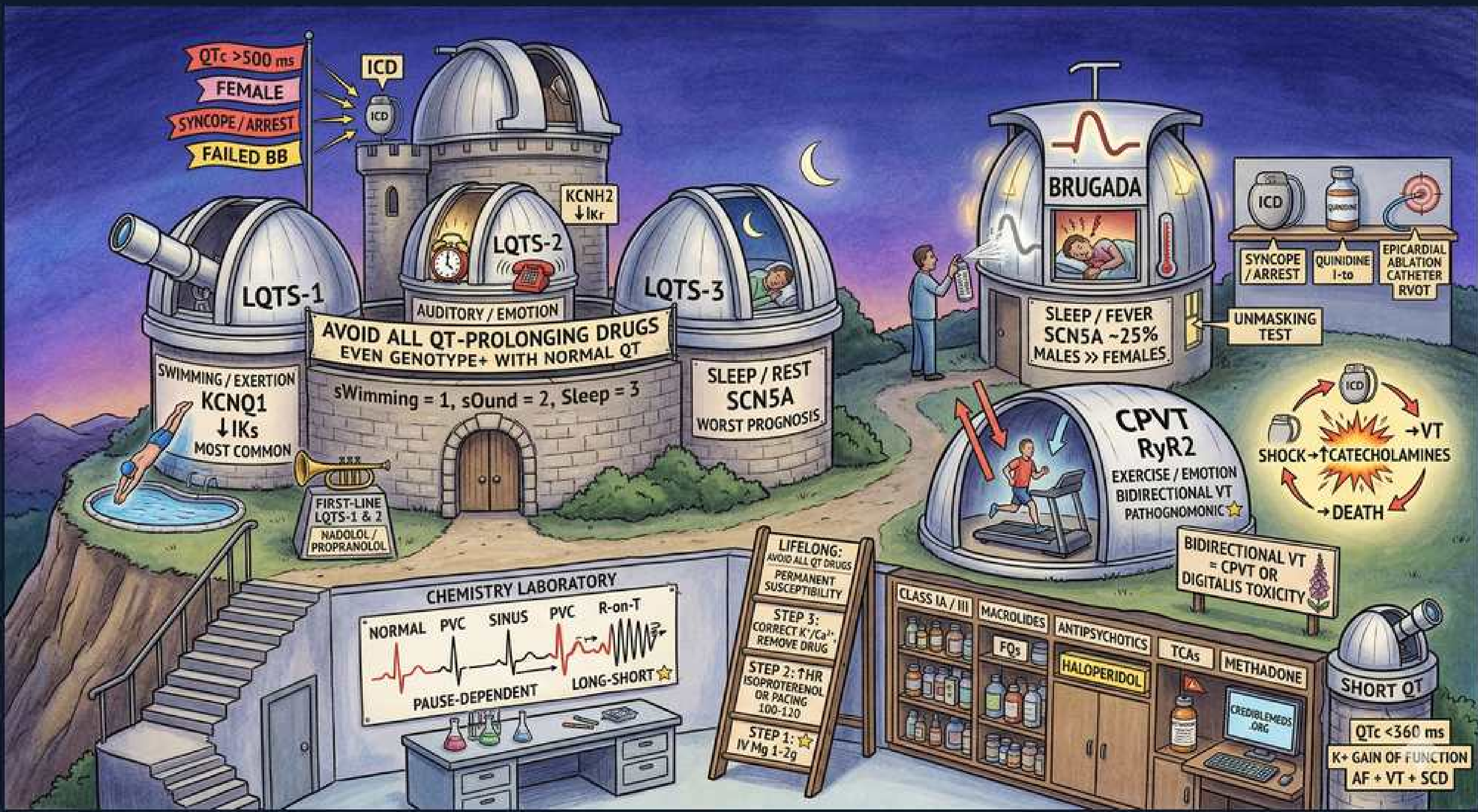
Arrhythmias — Inherited Channelopathies (LQTS, Brugada, CPVT, Short QT)



1. LQTS1 — exertion/swimming

WHAT YOU SEE

Observatory 'LQTS-1', 'KCNQ1 ↓IKs', 'SWIMMING/EXERTION', 'MOST COMMON'

WHAT IT ENCODES

Long QT syndrome type 1 is caused by a loss-of-function mutation in KCNQ1 (Kv7.1), reducing the slow delayed-rectifier potassium current IKs, which is needed to shorten the QT during sympathetic/exertional drive. It is the most common LQTS subtype (~30-35%), and events are triggered by exertion — classically SWIMMING/diving. The torsades risk rises because repolarization fails to keep pace with faster heart rates. Mnemonic: LQT-1 = 'one lap in the pool.'

BOARD TRAP

Attributing a swimming-related drowning/syncope in a young person to a simple accident is the trap — exertional or swimming-triggered syncope/arrest should raise LQTS-1; beta-blockers are highly protective here, and competitive swimming may need restriction.

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2. LQTS2 — auditory/startle

WHAT YOU SEE

'LQTS-2', 'KCNH2 ↓IKr', 'AUDITORY / STARTLE'

WHAT IT ENCODES

LQTS type 2 results from loss-of-function in KCNH2 (hERG), reducing the rapid delayed-rectifier potassium current IKr. It is the second most common subtype (~25-30%) and events are triggered by sudden AUDITORY/emotional startle — alarm clock, telephone ringing, doorbell. IKr is also the channel blocked by most drug-induced (acquired) long QT, so LQTS-2 carriers are exquisitely sensitive to QT-prolonging drugs. Mnemonic: LQT-2 = 'too loud' (auditory).

BOARD TRAP

Dismissing syncope provoked by a startling noise as vasovagal is the trap — auditory/startle-triggered syncope points to LQTS-2; postpartum is also a high-risk window for LQTS-2 events.

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3. LQTS3 — sleep/rest

WHAT YOU SEE

'LQTS-3', 'SCN5A', 'SLEEP / REST', 'WORST PROGNOSIS'

WHAT IT ENCODES

LQTS type 3 is a GAIN-of-function mutation in SCN5A, causing persistent (late) inward sodium current INa that prolongs repolarization. Events occur at REST or during SLEEP (bradycardia-dependent) and it carries the worst prognosis per event. Because the defect is excess sodium current, late-sodium-current blockers like mexiletine are particularly useful adjuncts, and beta-blockers are less protective than in LQTS 1/2. Note SCN5A LOSS of function instead causes Brugada — same gene, opposite direction.

BOARD TRAP

Assuming beta-blockers fully protect every LQTS patient is the trap — LQTS-3 (rest/sleep events) responds less to beta-blockade and benefits from mexiletine; the same SCN5A gene causes Brugada when it is loss-of-function, so direction of the mutation matters.

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4. QTc >500 = high risk

WHAT YOU SEE

Flags 'QTc >500 ms', 'FEMALE', 'SYNCOPE/ARREST', 'FAILED BB'

WHAT IT ENCODES

Markers of high sudden-cardiac-death risk in LQTS: QTc >500 ms (and especially >550 ms), prior syncope or aborted cardiac arrest, female sex (post-puberty), and breakthrough events despite beta-blockade. QTc is calculated by Bazett (QT/√RR); normal upper limits are ~450 ms (men) and ~470 ms (women). These features push toward ICD implantation.

BOARD TRAP

Relying on a single 'normal' QTc to exclude LQTS is the trap — QTc varies and genotype-positive patients can have intermittently normal QT; a borderline value plus syncope or family history of SCD still warrants treatment and genetic evaluation.

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5. Avoid all QT-prolonging drugs

WHAT YOU SEE

Banner 'AVOID ALL QT-PROLONGING DRUGS EVEN GENOTYPE+ WITH NORMAL QT'

WHAT IT ENCODES

All LQTS patients — including silent genotype-positive carriers with a normal resting QT — must avoid QT-prolonging drugs and correct hypokalemia/hypomagnesemia/hypocalcemia. Common offenders: antiarrhythmics (sotalol, dofetilide), macrolides/fluoroquinolones, azole antifungals, ondansetron, methadone, antipsychotics (haloperidol), and TCAs (see crediblemeds.org). Many act by blocking IKr (the hERG/KCNH2 channel).

BOARD TRAP

Giving ondansetron, azithromycin, or haloperidol to a known LQTS patient because the baseline QT 'looks fine' is the trap — these can precipitate torsades even in genotype-positive patients with normal resting QT; check the QT-prolonging drug list first.

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6. Beta-blocker first-line (LQTS)

WHAT YOU SEE

'FIRST-LINE LQTS 1 & 2 — NADOLOL / PROPRANOLOL'

WHAT IT ENCODES

Beta-blockers are first-line therapy for LQTS, blunting the sympathetic surge that triggers torsades; non-selective long-acting agents NADOLOL and PROPRANOLOL are preferred over metoprolol. They are MOST effective in LQTS-1 and LQTS-2 (adrenergically triggered). Left cardiac sympathetic denervation (LCSD) is an option for those with breakthrough events or who cannot tolerate beta-blockers.

BOARD TRAP

Choosing a pacemaker as first-line for LQTS is the trap — beta-blockade is first-line; pacing has only a niche role (e.g., pause-dependent LQTS-3 with bradycardia), and an ICD, not a pacemaker, is what protects against SCD in high-risk cases.

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7. ICD if arrest/refractory

WHAT YOU SEE
'ICD' icon near risk flags

WHAT IT ENCODES
An ICD is indicated in LQTS for survivors of aborted sudden cardiac death (secondary prevention, a Class I indication) and for patients with recurrent syncope or sustained ventricular arrhythmias despite adequate beta-blockade. Beta-blockers are continued alongside the ICD to reduce shock frequency.

BOARD TRAP
Implanting an ICD as the first step in every newly diagnosed asymptomatic LQTS patient is the trap — most are managed with beta-blockers and lifestyle/drug avoidance; the ICD is reserved for aborted arrest or beta-blocker-refractory events.

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8. Brugada — SCN5A

WHAT YOU SEE
'BRUGADA / SLEEP / FEVER / SCN5A ↓ / MALES >> FEMALES'

WHAT IT ENCODES
Brugada syndrome is most often a LOSS-of-function mutation in SCN5A (reduced sodium current INa), producing the type-1 pattern: coved ST-segment elevation ≥2 mm with T-wave inversion in right precordial leads V1-V2. VF/sudden death occurs at REST or during SLEEP and is more common in men of Southeast Asian descent (a cause of SUNDS — sudden unexpected nocturnal death). Same gene as LQTS-3 but opposite direction.

BOARD TRAP
Confusing the Brugada type-1 ST elevation in V1-V2 with an anteroseptal STEMI is the trap — Brugada has a coved 'shark-fin' morphology without reciprocal changes or troponin rise, in a stable patient; misreading it as MI sends them to the cath lab instead of EP evaluation.

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9. Fever/drugs unmask Brugada

WHAT YOU SEE
'UNMASKING TEST' / quinidine

WHAT IT ENCODES
Fever, sodium-channel-blocking drugs, vagal tone, and large meals/alcohol unmask or accentuate the type-1 Brugada ECG; a sodium-channel blocker (ajmaline, flecainide, or procainamide) challenge is used diagnostically to provoke the pattern. Therapy: aggressive fever control (antipyretics), ICD for those with prior arrest or syncope plus spontaneous type-1 pattern, and QUINIDINE (blocks transient outward current Ito) for electrical storm or as an ICD adjunct.

BOARD TRAP
Treating a Brugada VF storm with a class IC drug like flecainide is the trap — class IC sodium blockers WORSEN Brugada (they are used to unmask it); the correct storm therapy is isoproterenol acutely and quinidine, not a sodium-channel blocker.

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10. CPVT — exercise/emotion

WHAT YOU SEE
'CPVT / RyR2 / EXERCISE/EMOTION / BIDIRECTIONAL VT'

WHAT IT ENCODES
Catecholaminergic polymorphic VT (CPVT) is usually an autosomal-dominant gain-of-function mutation in the cardiac ryanodine receptor RyR2 (or recessive CASQ2), causing pathologic diastolic calcium leak from the sarcoplasmic reticulum under adrenergic stimulation. It presents in children/young adults with exercise- or emotion-induced syncope and a structurally normal heart with a NORMAL resting ECG. The hallmark arrhythmia is bidirectional VT (beat-to-beat 180° QRS axis alternation), which is essentially pathognomonic.

BOARD TRAP
Calling a normal resting ECG reassuring and missing CPVT is the trap — CPVT has a normal baseline ECG and normal echo, so exertional syncope with bidirectional/polymorphic VT on an exercise stress test (not resting ECG) makes the diagnosis; digoxin toxicity is the other classic cause of bidirectional VT.

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11. CPVT → catecholamines → death

WHAT YOU SEE
'CATECHOLAMINES → SHOCK → DEATH'

WHAT IT ENCODES
In CPVT, catecholamine surges directly drive the VT, so treatment is high-dose beta-blockers (NADOLOL preferred) often combined with FLECAINIDE (which here blocks RyR2-mediated calcium release and is beneficial), plus strict avoidance of competitive/strenuous sports. An ICD is added for survivors of arrest or breakthrough events — but always WITH a beta-blocker.

BOARD TRAP
Relying on ICD shocks ALONE to control CPVT is the trap — the pain/stress of a shock releases more catecholamines, triggering further VT and a vicious storm cycle; an ICD must always be paired with beta-blockade (± flecainide/sympathetic denervation), never used as monotherapy. Also note flecainide is good here, opposite to Brugada.

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12. Short QT

WHAT YOU SEE
'SHORT QT / QTc <360 ms / K+ GAIN OF FUNCTION / AF + VT + SCD'

WHAT IT ENCODES
Short-QT syndrome is a rare gain-of-function mutation in potassium channels (KCNH2/KCNQ1/KCNJ2) that accelerates repolarization, giving a QTc <360 ms (often <330) with tall peaked T waves. The abbreviated refractory period predisposes to atrial fibrillation (often in young patients), polymorphic VT, and sudden cardiac death. Quinidine can prolong the QT/refractory period; an ICD is the mainstay for those at high risk.

BOARD TRAP
Reading a very short QT as a benign normal variant is the trap — a markedly short QTc, especially with unexplained AF in a young person or a family history of SCD, signals short-QT syndrome and warrants ICD/EP evaluation, not reassurance.

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13. Pause/long-short → torsades

WHAT YOU SEE

Chemistry lab strip 'PVC SINUS PVC R-on-T / PAUSE-DEPENDENT / LONG-SHORT'

WHAT IT ENCODES

Torsades de pointes in long-QT physiology is classically pause-dependent and initiated by a 'long-short' sequence: a PVC, a compensatory pause that further prolongs the QT, then a second beat falling on the prolonged T wave (R-on-T) that triggers polymorphic VT twisting around the baseline. This is why pacing or magnesium (which stabilize and prevent pauses/EADs) help in acquired long QT.

BOARD TRAP

Reflexively shocking every torsades or reaching only for amiodarone is the trap — for torsades with a long QT, IV magnesium sulfate 1-2 g is first-line, and overdrive pacing or isoproterenol shortens the QT and abolishes pauses; QT-prolonging antiarrhythmics like amiodarone/sotalol can worsen it (defibrillate only if pulseless/sustained).